ARBI KULLAKSHI

Software Engineer · Rust & Robotics

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SUMMARY

Software engineer with 5+ years of experience building backend systems, security platforms, and delivery infrastructure. Develops autonomous robotics software in Rust with ROS 2, Gazebo, SLAM, state estimation, and path planning. Brings production-grade architecture and end-to-end ownership to robotics and SaaS systems.

PROFESSIONAL EXPERIENCE

Zennit **Jan 2024 – Present**

Full Stack Engineer

- Architected a nine-module Laravel SaaS backend and identity platform supporting 10 external OAuth providers, passwordless login, account linking, TOTP MFA, and auditable session termination.
- Designed and deployed `zennit/abac` as a production authorization layer, with mixed-policy evaluation over 10,000 resources completing in approximately 15 ms in controlled benchmarks.
- Delivered backend, frontend, and deployment systems across six products using Laravel, Django, React/Next.js, PostgreSQL, Redis, Docker, and Ansible.
- Automated containerized delivery and project setup with Docker, Traefik/Nginx, GitHub/GitLab CI, and reusable deployment workflows.

Hire A Loo **Aug 2025 – Jun 2026**

Backend Developer · Contract

- Authored 18 CRM PRs spanning search, web-lead ingestion, price quoting, email templating, CardDAV contacts, legacy migrations, and shared application infrastructure.
- Implemented a Typesense-powered global search experience with debounced queries, category pagination, persistent URL state, error handling, and automated backend coverage.
- Developed an end-to-end quote workflow with automatic status transitions, sanitized email templates, public accept/reject flows, previews, and feature tests.
- Authored the open-source `laravel-notifyre-sms` package with direct and notification-channel delivery, CLI/REST interfaces, webhook status tracking, optional persistence, and 183 tests.

Clickservice GmbH **Jan 2023 – Jan 2024**

Full Stack Developer

- Optimized Django and PostgreSQL services with targeted raw SQL where ORM-generated queries were inefficient.
- Implemented third-party integrations, automated reporting, and React/TypeScript interfaces for production workflows.

Vodafone Albania **Jan 2022 – Dec 2022**

Mobile Developer

- Delivered cross-platform React Native features for production iOS and Android applications.
- Integrated REST APIs, WebSockets, and authentication while contributing to automated release workflows.

EnRedYAte **Jan 2021 – Dec 2021**

Junior Frontend Developer

- Built responsive React and WordPress interfaces from product and design requirements.
- Contributed to design collaboration, application maintenance, and peer code review.

SELECTED OPEN SOURCE

- **`zennit/abac`** — Production-used Laravel 13 ABAC with policy constraints, middleware, caching, audit hooks, Artisan scaffolding, 61 tests, and ~15 ms benchmark evaluation at 10,000 resources.
- **`laravel-notifyre-sms`** — Notifyre SMS integration with notification channels, CLI/REST interfaces, delivery callbacks, persistence, and a 183-test suite.

TECHNICAL PROJECTS & RESEARCH

Cave Robot · Autonomous Cave-Navigation Drone **2026**

Rust · Serde · ROS 2 / `rclrs` · Gazebo · LiDAR · SLAM

- Delivered a working university robotics MVP as a six-crate Rust workspace spanning procedural generation, navigation, shared types, pathfinding, EKF localization, and SLAM.
- Implemented seeded 3D voxel caves and clearance-weighted A* outbound planning on the known map, while building a LiDAR/SLAM occupancy map during traversal for return-path planning.
- Integrated Rust with ROS 2 and Gazebo through Serde-based scan and command messages, translating LiDAR input into published velocity commands.

Roomba · Autonomous Cleaning Robot **2026**

Rust · Tokio · async-trait · nalgebra · Rerun · simulation-first architecture

- Delivered a documented 10× real-time simulation that autonomously visits reachable rooms, detects and cleans multiple dirt types, replans around obstacles, and returns to dock.
- Implemented timestamp-ordered sensor fusion with bounded replay for delayed encoder, IMU, beacon, and LiDAR measurements, including innovation-gated LiDAR-to-wall EKF corrections.
- Architected five independently clocked Tokio sensor tasks, fixed-step physics, and a sequential control/actuation boundary around clearance-aware occupancy-grid A* planning.

Event-Aware Traffic Prediction · B.Sc. Thesis **2026**

Python · scikit-learn · XGBoost · Caltrans PeMS

- Designed a reproducible study across 5.68 million sensor-time rows and 164 verified concerts, holding model type, rows, seeds, and forward test splits constant.
- Executed 540 controlled Random Forest and XGBoost fits with validation gates, placebo tests, bootstrap confidence intervals, benchmarks, and provenance tracking.
- Reported a negative real-data result: event metadata degraded predictions, while controlled synthetic experiments confirmed recovery of the injected event effect.

Pyramid Backstage · JunctionX Tirana Hackathon **2026**

TypeScript · Bun · Hono · Next.js · PostgreSQL · Drizzle · React Three Fiber

- Architected and delivered a full-stack venue-operations platform unifying event intake, space matching, quotations, reservations, approvals, and task planning.
- Built a typed monorepo with AI-assisted request processing, transactional availability checks, interactive 3D venue exploration, and Docker-based deployment.

TECHNICAL SKILLS

Programming Languages: Rust, PHP, Python, TypeScript

Robotics: ROS 2 / `rclrs`, Gazebo, LiDAR / SLAM, EKF / pathfinding

Rust Ecosystem: Tokio, Serde, Cargo, nalgebra

Frameworks: Laravel, Django, React / Next.js, React Native

Data and ML: PostgreSQL, Redis, Typesense, scikit-learn / XGBoost

Infrastructure: Docker, Traefik / Nginx, GitHub / GitLab CI, Ansible

EDUCATION & PROFESSIONAL DEVELOPMENT

Let's Get Rusty LLC **In progress · 2026**
Rust Live Accelerator · Virtual

Live advanced Rust program with a 100+ hour curriculum covering language features, design patterns, asynchronous services, persistence, embedded systems, and production delivery.

European University of Tirana **2026**
B.Sc. Informatics Engineering · Thesis: Event-Aware Traffic Prediction

Hermann Gmeiner High School **2023**
ICT Vocational Degree

LANGUAGES

Albanian (Native) · English (C2) · German (B1)